

Selection-Corrected Dynamical Likelihood for the Metallicity-Dependent Stellar Mass–Luminosity Relation from Gaia Wide Binaries

Selection-corrected dynamical likelihood: 1 October 2026

1 Selection-corrected dynamical likelihood

1.1 Dynamical likelihood of one pair

Write $s_{jq} = \sqrt{M_{\text{tot},jq}/M_{\odot}}$ for the mass scale of pair j at metallicity node Z_q , where $M_{\text{tot},jq}$ is the sum of the two component masses predicted by the MLR. The latent relative speed is $v = s \tilde{u}$, and the observed statistic u_j follows the Rice distribution

$$R(u_j | v, \sigma_{u,j}) = \frac{u_j}{\sigma_{u,j}^2} \exp\left[-\frac{u_j^2 + v^2}{2\sigma_{u,j}^2}\right] I_0\left(\frac{u_j v}{\sigma_{u,j}^2}\right), \quad u_j \geq 0, \quad (1)$$

where I_0 is the zeroth-order modified Bessel function (Rice, 1945). The good component has the mass-scaled density $p(\tilde{u})$ of the functional form of Hwang et al. (2024), with fixed constants (B, u_c, C) and support $0 < \tilde{u} \leq 80 \text{ km s}^{-1} \sqrt{\text{AU}}$. The outlier component is a Gaussian of mean 40 and standard deviation $13 \text{ km s}^{-1} \sqrt{\text{AU}}$ truncated to $[0, 80] \text{ km s}^{-1} \sqrt{\text{AU}}$, defined directly in the latent speed v with density $q(v)$, and therefore independent of the pair mass. The mixture likelihood of pair j is

$$\mathcal{L}_j(s) = (1 - f_{\text{outlier}}) \int_0^{80} p(\tilde{u}) R(u_j | s\tilde{u}, \sigma_{u,j}) d\tilde{u} + f_{\text{outlier}} \int_0^{80} q(v) R(u_j | v, \sigma_{u,j}) dv. \quad (2)$$

The first integral carries the Jacobian $1/s$ of the change of variables from $\tilde{u}$ to v .

1.2 Truncation by the significance cut

The catalogue retains only pairs whose proper-motion difference is detected at more than $c = 3$ times its uncertainty. Because $\sigma_{u,j} = u_j/(\Delta\mu/\sigma_{\Delta\mu})_j$, this cut is a threshold on the observed statistic,

$$u_j > c \sigma_{u,j}, \quad (3)$$

with $\sigma_{u,j}$ fixed for each pair by the catalogue astrometric uncertainties, the parallax and the projected separation. Pairs with small latent speed and large $\sigma_{u,j}$ are preferentially removed, so the observed u_j are not a draw from equation (2). The density of u_j conditional on passing the cut is

$$\mathcal{L}_j^{\text{sel}}(s) = \frac{\mathcal{L}_j(s)}{\Pi_j(s)}, \quad (4)$$

where $\Pi_j(s)$ is the probability that a pair with mass scale s satisfies equation (3).

For a latent speed v , the Rice distribution gives the probability of passing the cut as the Marcum Q function,

$$\Pr(u > c \sigma_{u,j} | v) = Q_1\left(\frac{v}{\sigma_{u,j}}, c\right). \quad (5)$$

Averaging over the two mixture components,

$$\Pi_j(s) = (1 - f_{\text{outlier}}) \Pi_j^{\text{good}}(s) + f_{\text{outlier}} \Pi_j^{\text{out}}, \quad (6)$$

with

$$\Pi_j^{\text{good}}(s) = \int_0^{80} p(\tilde{u}) Q_1\left(\frac{s\tilde{u}}{\sigma_{u,j}}, c\right) d\tilde{u}, \quad \Pi_j^{\text{out}} = \int_0^{80} q(v) Q_1\left(\frac{v}{\sigma_{u,j}}, c\right) dv. \quad (7)$$

The pass probability of the outlier component does not depend on the mass, and Π_j^{good} depends on the mass and the uncertainty only through $s/\sigma_{u,j}$. It increases with s and tends to one when $s\tilde{u} \gg c\sigma_{u,j}$ over the support of $p(\tilde{u})$. Dividing by Π_j therefore removes the probability that the untruncated model places below the cut, where no pair is observed; this probability is larger at low mass, so the correction lowers the inferred masses relative to [equation \(2\)](#).

Metallicity is marginalized with the Stage-1b weights. Selection acts on the true mass of the pair, so the factor Π_j is evaluated at each node before the sum,

$$\mathcal{L}_j^{\text{sel}}(\Theta) = \sum_q \bar{P}_{jq} \frac{\mathcal{L}_j(s_{jq})}{\Pi_j(s_{jq})}. \quad (8)$$

The dynamical term of the log-posterior is $\sum_j \ln \mathcal{L}_j^{\text{sel}}$. The fraction of outliers in the selected sample is $f_{\text{outlier}} \Pi_j^{\text{out}} / \Pi_j$, which differs from f_{outlier} ; the parameter f_{outlier} is the mixing fraction before selection.

1.3 Role of the fixed velocity distribution

The function $p(\tilde{u})$ describes the distribution of the mass-scaled statistic before selection, determined by eccentricity, orbital phase and orientation. With the cut modelled in [equation \(4\)](#), the shape is held at the fixed constants and is not adjusted to absorb the selection. Its upper edge is set by energy conservation: for a bound orbit $u^2 \leq v^2 r < 2Gm_{\text{tot}}$, hence

$$\tilde{u} < \sqrt{2GM_{\odot}/\text{AU}} = 42.12 \text{ km s}^{-1} \sqrt{\text{AU}}, \quad (9)$$

independently of the eccentricity distribution. This edge fixes the mass scale of the velocity distribution.

1.4 Numerical evaluation

The function Π_j^{good} is tabulated once as a function of $x = s/\sigma_{u,j}$ on a logarithmic grid, using trapezoidal integration over $\tilde{u}$ and the survival function of the non-central χ^2 distribution with two degrees of freedom, $Q_1(a, c) = \Pr[\chi_2^2(a^2) > c^2]$. Values of $\ln \Pi_j^{\text{good}}$ at the mass nodes of the dynamics lookup are obtained by linear interpolation in $\ln x$. The quantity Π_j^{out} is computed once per pair. The tables reproduce Monte Carlo pass fractions of the two components to within 4×10^{-3} . The correction is enabled with the option `-selection-cut`, which sets c .

1.5 Assumptions of the selection model

Only the threshold of [equation \(3\)](#) is modelled. The remaining catalogue criteria (chance-alignment probability, fidelity, parallax, photometric signal-to-noise and astrometric quality cuts) are assumed not to depend on u_j at fixed mass. The uncertainty $\sigma_{u,j}$ is treated as fixed for each pair. The Stage-1b metallicity posteriors $\bar{P}_{jq}$ are computed from the selected sample and carry no selection correction. The pass probability is evaluated with the same shape of $p(\tilde{u})$ and outlier component as the likelihood, so any mismatch between these fixed shapes and the true population affects Π_j as well as $\mathcal{L}_j$.

References

- Hsiang-Chih Hwang, Yuan-Sen Ting, Sihao Cheng, and Joshua S. Speagle. Dynamical masses across the Hertzsprung–Russell diagram. *Monthly Notices of the Royal Astronomical Society*, 528(3):4272–4288, 2024. doi: 10.1093/mnras/stae297.
- Stephen O. Rice. Mathematical analysis of random noise. *Bell System Technical Journal*, 24(1): 46–156, 1945. doi: 10.1002/j.1538-7305.1945.tb00453.x.